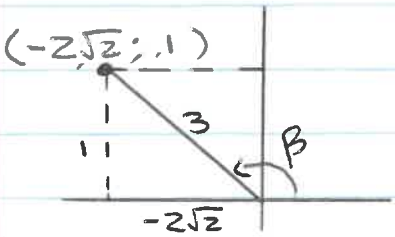
Grade 12 Cycle Test 29 April

Memo

Question 1

1.1.



$x^2 = (3)^2 - (1)^2$

$x^2 = 8$

$x = \pm\sqrt{8} = \pm 2\sqrt{2}$

✓Quad

✓y & r ✓x

$\tan\beta = -\dfrac{1}{2\sqrt{2}}$ ✓CA

(4)

1.2. $\cos 2\beta$

$= 1 - 2\sin^2\beta$ ✓

$= 1 - 2\left(\dfrac{1}{3}\right)^2$ ✓

$= 1 - \dfrac{2}{9}$

$= \dfrac{7}{9}$ ✓

(3)

Question 2

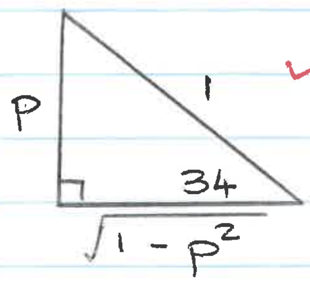
2.1. $\sin 214$

$= -\sin 34$ ✓

$= -p$ ✓ (2)

2.2. $\sin 68$

$= 2\sin 34 \cos 34$ ✓



✓m

$= 2p\sqrt{1-p^2}$ ✓ (3)

2.3. $\cos 26$

$= \cos(60-34)$ ✓

$= \cos 60 \cos 34 + \sin 60 \sin 34$ ✓

$= \frac{1}{2}\sqrt{1-p^2} + \frac{\sqrt{3}}{2} p$ ✓ (3)

Question 3

3.1 $$\frac{\sin x}{\sin 2x} + \frac{\sin(180-x)}{\cos x} - \frac{\sin^2 x + \cos^2(180+x)}{2\sin(90+x)}$$

$$= \frac{\sin x}{2\sin x\cos x} + \frac{\sin x}{\cos x} - \frac{\sin^2 x + \cos^2 x}{2\cos x}$$

$$= \frac{1}{2\cos x} + \frac{\sin x}{\cos x} - \frac{1}{2\cos x}$$ (1)

$$= \frac{\sin x}{\cos x}$$

$= \tan x$ (6)

3.2. $$\frac{\sin 53 \cos 11 - \cos 53 \sin 11}{\sin 21 \cos 21}$$

$$= \frac{\sin(53-11)}{\sin 21 \cos 21}$$

$$= \frac{\sin 42}{\sin 21 \cos 21} \times \frac{2}{2}$$ m

$$= \frac{2\sin 42}{\sin 42}$$

$= 2$ (4)

Question 4

4.1. $$\frac{\sin x + \sin 2x}{1 + \cos x + \cos 2x} = \tan x$$

$$LHS = \frac{\sin x + 2\sin x\cos x}{1 + \cos x + (2\cos^2 x - 1)}$$

$$= \frac{\sin x + 2\sin x \cos x}{\cos x + 2\cos^2 x}$$

$$= \frac{\sin x\,(1 + 2\cos x)}{\cos x\,(1 + 2\cos x)}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

$\therefore$ LHS = RHS (5)

4.2.1. $4\cos\theta - \dfrac{\cos 2\theta + 1}{\cos\theta} = \dfrac{\sin 2\theta}{\sin\theta}$

$\text{LHS} = 4\cos\theta - \dfrac{2\cos^2\theta - 1 + 1}{\cos\theta}$ ✓

$= \dfrac{4\cos^2\theta - 2\cos^2\theta}{\cos\theta}$ ✓

$= \dfrac{2\cos^2\theta}{\cos\theta}$ ✓

$= 2\cos\theta$ ✓ (4)

4.2.2. $\cos\theta = 0$

$\underbrace{\cos\theta = 0 \text{ and } \sin\theta = 0}$ ✓

$\theta = 90^\circ + k \cdot 180^\circ$

$\sin\theta = 0$ ✓

$\theta = k \cdot 180^\circ$

$\therefore \theta \in \{180; 270; 360\}$ ✓ (3)

Question 5

$3\sin\theta\sin 22 = 3\cos\theta\cos 22 + 1$

$3\sin\theta\sin 22 - 3\cos\theta\cos 22 = 1$

$-3(\cos\theta\cos 22 - \sin\theta\sin 22) = 1$

$\cos(\theta + 22) = -\frac{1}{3}$

ref $\angle = 70,53$

Quad 2

$\theta + 22 = 109,47 + k \cdot 360$

$\theta = 87,47 + k \cdot 360 \quad ; k \in \mathbb{Z}.$

or

Quad 4

$\theta + 22 = 289,47 + k \cdot 360$

$\theta = 267,47 + k \cdot 360 \quad ; \quad k \in \mathbb{Z}.$

(6)

Question 6

$$\frac{\cos 2\theta}{\sin(\theta - 45)}$$

$$= \frac{\cos^2\theta - \sin^2\theta}{\sin\theta\cos 45 - \cos\theta \sin 45}$$ ✓ ✓

$$= \frac{(\cos\theta + \sin\theta)(\cos\theta - \sin\theta)}{-\frac{\sqrt{2}}{2}(\cos\theta - \sin\theta)}$$ ✓HCF ✓special ∠

$$= T \div -\frac{\sqrt{2}}{2}$$ ✓subs

$$= -\frac{2T}{\sqrt{2}}$$ ✓

(6)